

Donnell Rodriguez

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Independent Software Engineer | Reinforcement Learning Engineer
AI Agent Engineer and Medical AI Inference Deployment Contractor

Experience

Independent Software Engineer / Reinforcement Learning Engineer
Remote / China
2024 - Present

I work as an independent software engineer focused on reinforcement learning systems, AI agent infrastructure, open-source framework maintenance, and production-oriented AI model deployment.

My work spans RL training frameworks, RLHF tooling, agent runtime systems, Linux GPU server deployment, FastAPI backend integration, React frontend delivery, Nginx reverse proxy configuration, PostgreSQL/RabbitMQ service integration, and DICOM/NIfTI medical imaging workflows.

Public works and engineering contributions include:

- EnvPool: worked on high-performance reinforcement learning environment execution and related engineering workflows.
- OpenRLHF: contributed to RLHF training framework usage, maintenance, and deployment-oriented integration work.
- TRL: worked with Hugging Face TRL-style reinforcement learning fine-tuning workflows and related framework maintenance.
- OpenHands: participated in remote open-source AI agent development and maintenance with US-based collaborators.
- OpenClaw / OpenClaw Context Compactor: maintained AI agent context compaction and structured state tooling for longer, more stable agent runs.
- Medical AI Inference Deployment: deployed and integrated medical image inference systems involving DICOM upload, NIfTI conversion, 3D result visualization, and doctor-side NIfTI review package download.

Representative engineering responsibilities:

- Built and maintained reinforcement learning and RLHF workflow components.
- Worked with open-source RL frameworks including EnvPool, OpenRLHF, and TRL.
- Maintained AI agent infrastructure and context-management tooling.
- Deployed medical AI inference frontend and backend services on Linux GPU servers.
- Configured Nginx reverse proxy for public domain, local machine, and future hospital LAN access.
- Integrated React/TanStack frontend workflows with FastAPI backend APIs.
- Supported PostgreSQL-backed patient, task, file, and result workflows.
- Supported RabbitMQ-based serial inference task queue design for GPU workload control.
- Worked with DICOM upload, DICOM-to-NIfTI conversion, NIfTI candidate selection, nnUNet-style inference, segmentation masks, and 3D result viewer delivery.
- Added doctor-facing NIfTI review package download workflow for manual verification and correction.

- Updated frontend deployment strategy to use same-origin API requests instead of hardcoded server IPs.

Selected Open-Source / Public Works

🐙 EnvPool: A Highly Parallel Reinforcement Learning Environment Execution Engine
GitHub: <https://github.com/sail-sg/envpool>

EnvPool is a high-performance parallel environment executor for reinforcement learning workloads. It is designed to accelerate RL agent training by executing environments efficiently through C++ thread pools, pybind11 integration, and vectorized environment interfaces.

My related work focuses on understanding, maintaining, and applying high-throughput RL environment execution patterns for reinforcement learning training and deployment workflows.

🐙 OpenRLHF
GitHub: <https://github.com/OpenRLHF/OpenRLHF>

OpenRLHF is an open-source RLHF framework for post-training and alignment of large language models. My work involves framework maintenance, usage, integration, and engineering support around RLHF-style training pipelines.

Related areas:

- RLHF training workflow maintenance
- PPO / reward model / policy optimization workflow understanding
- Distributed training and deployment-oriented integration
- Engineering support for model post-training workflows

🐙 TRL
GitHub: <https://github.com/huggingface/trl>

TRL is a reinforcement learning and fine-tuning library for transformer language models. My related work includes maintaining and working with TRL-style training workflows for model alignment and reinforcement learning fine-tuning.

Related areas:

- RL fine-tuning workflow engineering
- Transformer model post-training workflows
- Alignment pipeline integration
- Practical experimentation and maintenance support

🐙 OpenHands
GitHub: <https://github.com/All-Hands-AI/OpenHands>

OpenHands is an open-source AI agent platform for software engineering automation. I participated in remote open-source maintenance and AI agent engineering work, including agent runtime workflows and integration-oriented development.

Related areas:

- AI agent development
- Remote open-source collaboration
- Agent workflow maintenance
- Tool-use and software engineering automation systems

🐙 OpenClaw / OpenClaw Context Compactor

GitHub: <https://github.com/donnell-rodriguez/openclaw-context-compactor>

OpenClaw Context Compactor is a tool for compacting AI agent conversation context into structured state for smaller prompts, lower token usage, and more stable agent runs.

Related areas:

- AI agent context management
- Prompt and memory compaction
- Structured state representation
- Long-running agent stability

Selected Project

Medical Imaging AI Inference System Deployment

A server deployment and integration project for a medical imaging AI workflow. The system supports DICOM upload, NIfTI conversion, inference job creation, 3D result visualization, and doctor-side NIfTI review package download.

Scope of work:

- Linux GPU server deployment
- Frontend static build and Nginx release deployment
- FastAPI backend integration verification
- Public domain access configuration
- Local Linux desktop access through `http://127.0.0.1/`
- Same-origin API configuration for domain, local, and future LAN access
- DICOM/NIfTI workflow support
- RabbitMQ serial inference queue support
- 3D medical result viewer workflow adjustment
- Deployment verification and customer handoff preparation

Outcome:

- Public demo access through HTTPS domain
- Local machine access for hospital-side usage
- Production frontend build no longer tied to fixed server IP
- Prepared architecture for future hospital LAN access
- Improved doctor-facing workflow for result review and NIfTI package download

Education

Dalian University of Technology	Dalian, China
Master of Computer Science	2022 - 2025
Bachelor of Computer Science	2018 - 2022

Technical Skills

Reinforcement Learning:

RL systems, RLHF workflows, PPO-style training, reward modeling workflows, model alignment, high-throughput environment execution

AI Engineering:

Model inference deployment, LLM post-training workflow integration, AI agent systems, tool-use agents, context compaction

Backend:

Python, FastAPI, REST APIs, PostgreSQL, RabbitMQ, background workers, authentication, protected file access

Frontend:

React, TanStack, JavaScript, clinical workflow UI integration, 3D viewer integration

Deployment:

Linux GPU servers, Nginx, domain configuration, same-origin API deployment, systemd, GitHub-based delivery workflow

Medical Imaging:

DICOM upload, DICOM ZIP/folder workflows, DICOM-to-NIfTI conversion, NIfTI candidate selection, segmentation result delivery, 3D medical viewer deployment

Engineering Focus

I specialize in practical deployment work between AI research code and real customer-facing systems. My role is not limited to writing isolated scripts; I help connect models, backend APIs, frontend workflows, server configuration, task queues, file storage, and deployment verification into a working system that researchers, doctors, and customers can actually use.

Disclaimer

Independent contractor. Not an employee of any university, hospital, or medical institution unless separately stated in a written contract.